

Exhaustivity and nominal predication in the left periphery

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1 Introduction

The clause is commonly seen as consisting of several structural layers, each with different properties. For example, Rizzi (1997) talks about (i) the *lexical* layer, where θ -roles are assigned, (ii) the *inflectional* layer, where argumental features such as case and agreement are licensed, and (iii) the *complementizer* layer—a.k.a. the left periphery—which hosts topics and various operator-like elements, such as interrogative and relative pronouns, focused elements, and so on. This observation is also captured in Grohmann’s (2003) clausal tripartition into “prolific domains”: the θ – domain, the φ – domain, and the ω – domain, the last of which Grohmann defines as part of the derivation where discourse information is established. Rizzi (1997) views the left periphery as the interface between a propositional content expressed by the IP, and a superordinate structure (a higher clause, or the articulation of discourse). Crucially, he does not consider the left periphery to be part of the extended projection of the verb, but a layer fundamentally distinct from the IP. For Rizzi, the *topic and focus field* in the left periphery expresses the “topic-comment and focus-presupposition articulations”, respectively, and elements that occupy the left periphery are endowed with [Topic] and [Focus] features. Most subsequent work, especially in the cartographic tradition, has also assumed that the left periphery encodes primarily information-structural content of the utterance, though it has also been noted that the left-peripheral ‘focus’ position appears to be, at least in some languages, tied not to discourse prominence in general, but to exhaustivity, a property such structures share with English-style clefts (see, e.g., É. Kiss 1998; Horvath 2007 on Hungarian).

This paper investigates a phenomenon found in some languages with a rich left periphery (so-called *discourse-configurational* languages), where nominal predicates in discourse-neutral contexts occur in apparently the same position that is otherwise occupied by focused/exhaustified constituents. In the West Atlantic Niger-Congo language Wolof, one of the languages investigated in this paper, wh-words and exhaustified constituents A'-move to the specifier of the A'-movement complementizer (*la*), as in (1). Nominal predicates are found in the same position, shown in (2). I here refer to clauses with nominal predicates as *copular clauses* for simplicity, even when sentences at the center of our investigation do not have a verbal copula.

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| <p>(1) <i>Exhaustive identification</i>¹</p> <p>Gato l-a=ñu sàcc.
 cake l-C_{WH}=3PL steal
 <i>'It's cake that they stole.'</i></p> | <p>(2) <i>Nominal predication</i></p> <p>Sàcckat l-a=ñu.
 thief l-C_{WH}=3PL
 <i>'They are thieves.'</i></p> |
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Due probably to an Indo-European bias, both the descriptive, and the formal literature, commonly describe such clauses, especially when it comes to African languages, in the following way. Elements such as *(l)a* in Wolof, that I consider a complementizer, are termed ‘copulas’, even if they do not display any verbal characteristics, such as the ability to carry inflectional morphology, like tense or aspect marking, or be modified by an adverb. Additionally, many of these languages do have copulas which do exhibit verbal properties, as the Wolof example with subject exhaustification in (3) shows.²

- (3) *Nominal predication with a copula*
- Xale yi a (>yee) **di**-oon sàcckat.
 child the.PL C_{WH} COP-PST thief
'It's the children who were thieves.'

If *(l)a* in (1) is analyzed as a verbal copula, then it is argued that this sentence is a bi-clausal cleft. This requires positing a null expletive subject, and the remainder of the clause is treated as a relative clause; there is clear evidence against this view in Wolof. The motivation for this approach also comes from the interpretation of sentences such as (1), which, as the translation indicates, is close to that of English clefts, and is partially probably due to the fact that sentences as in (2) have the same element *(l)a*, which, through the lense of Indo-European languages, then must be similar to *be*.

In this paper, I discuss A'-fronting for exhaustivity/focus and copular clauses in two typologically distinct languages: Wolof, previewed above, and Igala, a Yoruboid language spoken in Nigeria. We shall see that there is solid morphosyntactic evidence supporting the claim that structures as in (1) are monoclausal in Wolof. In Igala, on the other hand, such constructions have mixed mono-clausal and bi-clausal properties, suggesting a possible ongoing grammaticalization process. Crucially, clauses with nominal predicates as in (2) can be formed in the left periphery in both languages, without being accompanied by the typical information-structural effects associated with left-peripheral elements or cleft constructions. I discuss the repercussions of these findings for our understanding of the left periphery in the architecture of the grammar, and propose that the prevalent view, according to which it is primarily the locus of discourse-related information, is at the very least incomplete. I propose that languages such as Wolof and Igala, where nominal predication and fronting for exhaustivity/focus overlap are the key to furthering our understanding of the left periphery and its relationship with the extended projection of the verb. My primary goal here is to describe the empirical landscape, and offer some suggestions as to what an analysis should be able to explain; I do not offer a new account of the left periphery at this point.

The paper is organized as follows. In section 2 I briefly address a few relevant examples of clefts and cleft-like constructions, focusing on African languages, and the morphosyntactic similarities between those structures and copular clauses. Sections 3 and 4 lay out the proper-

¹Abbreviations follow the Leipzig Glossing Rules, with the addition of EMB = embedded, PP = pre-prefix (augment), PRON = pronoun, SFP = sentence-final particle, STR = strong pronoun.

²The 3rd plural morpheme *ñu* in (2) is a clitic pronoun, not agreement (Martinović 2015, 2023a).

ties of the left periphery in Wolof and Igala, respectively, and give evidence that clauses with nominal predicates in both languages involve A'-movement of the nominal predicate and the topicalization of the subject. Section 5 discusses the issues that the phenomenon presents us with, considers existing analyses, and lays out an informal proposal for what a comprehensive analysis ultimately needs to account for.

2 Clefts in Niger-Congo

Clefts are cross-linguistically a common structure, usually described as separating the ‘discourse prominent’ constituent from the background; that is, they are considered to be tied to the expression of ‘focus’ (the term is generally used fairly informally and with inconsistent meaning in the literature). I here use *cleft* to designate biclausal structures similar to English clefts, involving a copular clause hosting the clefted constituent, and a constituent that has properties of a relative clause, as in (4).

- (4) *English cleft construction*
It was **beans** [that Ben cooked].

We have evidence that, in English, the clefted constituent, *beans*, arrives at its surface position via A'-movement, but there is no agreement on what the structure of English clefts is (for a summary of various proposals, see, e.g., Hartmann and Veenstra 2013).

Across Niger-Congo, cleft-like constructions involving A'-movement are common. In some languages, the structures in question indeed look very similar to English-style clefts, as the example in (5) from Lubukusu (Bantu) shows: there is a copular element (*ba*) which exhibits agreement, and a complementizer-like element that is also found in (some) relative clauses (*ni*).

- (5) *Object cleft in Lubukusu*
Ka-ba ka-ma-kaanda ni-k-o Nangila a-a-tekha.
6-be PP-6-beans NI-6-PRON 1Nangila 1-PST-cook
‘It was beans that Nangila cooked.’ (Wasike 2006: 144)

There are also languages with A'-movement in structures expressing focus/exhaustivity where we do not appear to be dealing with biclausal cleft constructions. This is the case in Hausa (Afro-Asiatic), which has an element that occurs with fronted constituents, *nè* in (6), variably analyzed as either a left peripheral head (Green 2007), or a focus-sensitive exhaustivity marker (Hartmann and Zimmermann 2007). (Throughout the paper, I use the English cleft as the translation of all the constructions that have been argued to encode exhaustivity for simplicity and clarity, even if they do not involve bi-clausal clefting.)

- (6) *Object exhaustivity in Hausa*
Abinci₁ (**nè**) yârā sukà sà yā t₁
food NE children 3PL.FOC.PFV buy
‘It is food that the children bought.’ (Green 2007)

Green (2007) shows that sentences as in (6) involve A'-movement to a left-peripheral position, and argues that there is no morphosyntactic evidence that would justify treating them as bi-clausal clefts of the English type.

Then there are languages that seem to be somewhere in-between. Gatchalian (2023, to appear) investigates cleft constructions in Kirundi (Great Lakes Bantu), which contain the same element *ni* found in Lubukusu and many other Bantu languages, with the crucial difference that it precedes the clefted constituent, as in the object cleft in (7), making it look less like a head to whose specifier A'-extracted elements move.

(7) *Object cleft in Kirundi*

Ni igitabu Yohaáni a-a-som-yé.

NI 7book Yohani 1SM-PST-read-PFV.EMB

'It is the book that Yohani read.' (Gatchalian to appear: 1)

Gatchalian argues that clefts in Kirundi are biclausal, relying on various morphosyntactic properties of the putative relative clause in (7), which align with those of embedded, and not matrix, clauses: embedded verbs occur with a distinct tone melody, negation differs between embedded and matrix clauses, and the conjoint/disjoint alternation is available only in matrix clauses. According to all these properties, the part of the clause following the clefted element *igitabu* 'book' behaves like an embedded clause.

The view that structures as above are parallel to English clefts also comes from the fact that the same morphology that occurs in focus/exhaustivity constructions is commonly found in copular sentences in these languages. Examples (8)-(9) show copular sentences in Kirundi and Hausa, with the elements *ni/nè*.

(8) *Copular clause in Kirundi*

Yohaani **ni** umunyeshuúre.

John NI 1.student

'John is a student.'

(9) *Copular clause in Hausa*

Audù dālibī **nè**

Audu student.M NE

'Audu is a student.'

Kirundi sentences as in (8) cannot contain any inflectional morphology or agreement, and they can only be used for 3rd person singular subjects; a verbal copula is used in all other cases. Therefore, even if Kirundi clefts are bi-clausal, in the sense that the clause following the clefted element in (7) behaves like an embedded clause, the matrix 'clause' is not a full, finite clause. This is essentially what Gatchalian (2023) argues for in the end – that *ni* is a non-verbal head, Pred, and that it does not have an extended projection, but forms some sort of a reduced clause; I return to this in more detail in section 5. Green (2007), on the other hand, argues that focus/exhaustivity constructions are monoclausal in Hausa, and proposes that copular clauses as in (9) involve A'-movement of the predicate to the left periphery and the topicalization of the subject. In the following two sections, I argue that copular clauses in Wolof and Igala are built in the same way.

3 The left periphery and copular clauses in Wolof

In this section, we first familiarize ourselves with the relevant aspects of Wolof clause-structure and the properties of its left periphery, and then look at how copular clauses pattern with respect to other structures in Wolof. The main goal of this section is to establish that Wolof copular clauses are built in the left periphery, with the subject being a syntactic topic, co-indexed with a clause-internal clitic, and the predicate A'-moving to Spec,CP.

3.1 The structure of the left periphery in Wolof

Finite clauses in Wolof are built fairly ‘high’; they all have an overt complementizer layer hosting a variety of so-called *sentence particles* (e.g., Dunigan 1994; Martinović 2015, 2023a).³ In finite indicative clauses in non-extraction contexts, a non-clitic subject is obligatorily in a left-peripheral position, and co-indexed with a clitic pronoun clause-internally, as in (10) and (11). The non-clitic and the clitic subject can never occupy the same syntactic position in the clause, and in these structures, the non-clitic subject is optional, only the clitic is obligatory.

(10) *Neutral clause*

Ñaari say-say₁ taal na=*(ñu₁) kër-u Sidi.
 two criminal burn.down C=3PL house-GEN Sidy
 ‘Two criminals burned down Sidy’s house.’

(11) *Predicate focus clause*

Ñaari say-say₁ da=*(ñu₁) taal kër-u Sidi.
 two criminal C=3PL burn.down house-GEN Sidy
 ‘Two criminals burned down Sidy’s house.’

Evidence that the clitic in (10)-(11) is not agreement comes from A’-extraction constructions, where the clitic and the non-clitic subject occur in complementary distribution clause-internally (below C), as shown in (12).

(12) *Non-subject extraction*

- a. Lan₁ l-a {say-say yi/ñu} taal t₁?
 what l-C_{WH} criminal the.PL/3PL burn.down
 ‘What did the criminals/they burn down?’
- b. Kër-u Musaa₁ l-a {say-say yi/ñu} taal t₁.
 house-GEN Moussa l-C_{WH} criminal the.PL/3PL burn.down
 ‘It’s Moussa’s house that the criminals/they burned down.’

Either the clitic or the non-clitic subject are obligatory, showing that the fact that the non-clitic can be omitted in non-extraction contexts does not indicate that the language is *pro*-drop, but is readily explained if the non-clitic subject is in a left-peripheral position; for extensive argumentation and more details on the differences between non-extraction and extraction clauses, see Martinović (2015, 2023a). Note that subjects commonly occur in left-peripheral positions in finite indicative non-extraction/matrix clauses across the Niger-Congo language family (a.o., Baker 2003; Henderson 2006; Schneider-Zioga 2000, 2007; Pietraszko 2021).

A’-constructions come in several guises, their surface morphosyntax differing in various sentence-types (see Dunigan 1994; Torrence 2012, 2013a,b,c; Martinović 2015, 2017, 2023a). We are here interested only in the constructions featuring the element (*l*)*a*, as in (12), argued in Martinović 2015, 2017 to be an A’-movement complementizer. The complementizer exhibits a subject/non-subject asymmetry (Martinović 2017), surfacing as *la* in the extraction of a non-subject, and as *a* in subject extraction, shown in (13).

³The data presented in this section have been collected from native Wolof speakers in St.-Louis, Senegal through primary fieldwork conducted by the author, on field trips between 2014 and 2024. The data have been repeatedly tested and confirmed by over a dozen speakers.

(13) *Subject extraction*

- a. Kan₁ a t₁ taal kër gi?
 who C_{WH} burn.down house the
 ‘Who burned down the house?’
- b. Ngañ₁ a t₁ taal kër gi.
 Ngañ C_{WH} burn.down house the
 ‘It’s Ngañ that burned down the house.’

It also common for a subject pronoun to surface in front of *a* in subject extraction, making (14) parallel to (13a).

(14) *Subject extraction*

- Kan mu-a (>moo) taal kër gi?
 who 3SG-C_{WH} burn.down house the
 ‘Who burned the house?’

There is no evidence that clauses in (14) are structurally different from clauses in (13): both constructions pass the standard movement tests, topicalized constituents occur before the question word in both cases, and nothing can intervene between the question word and C or the pronoun-C complex. Structures as in (14) possibly involve agreement between the element in Spec,CP and C; I leave this issue aside here, as all speakers I consulted use *a* and pronoun-*a* interchangeably in all contexts. I use both types of examples in the paper.

Torrence (2013a) shows that structures with (*l*)*a* pass the majority of the standard movement tests: sensitivity to strong and weak islands, reconstruction effects (idioms, Principle A, Principle C, quantifier binding), and strong crossover.⁴ They do not have properties of bi-clausal clefts: there is no pronominal subject (and there would obligatorily need to be one, as Wolof is not a *pro*-drop language, as we saw above), and there is also no relative complementizer, which is otherwise obligatorily overt in Wolof (see Torrence 2005, 2012, 2013c; Martinović 2015, 2017); compare (15) and (16).⁵

(15) *Relative clause*

- nit *(ku) Musaa gis
 person C_{REL} Moussa see
 ‘a person that Moussa saw’

(16) (*l*)*a* question

- Kan l-a (*ku) Musaa gis?
 who l-C_{WH} C_{REL} Moussa see
 ‘Who did Moussa see?’

Additionally, no inflectional morphology can occur in the supposed copular clause of the cleft. There is no way to say something like *Who was it that Moussa saw?*, for example. Wolof has one optional past tense morpheme, (*w*)oon (Bochnak and Martinović 2019), which is usually affixed onto the verb, but it can also occur in clauses with no verbal element, as is the case in copular clauses, as we shall see below. In clauses with A'-fronting, it can only be suffixed onto the lexical verb, and cannot occur in the higher part of the clause.

⁴Like many other languages, Wolof does not exhibit weak crossover; see Torrence 2013a for a detailed discussion.

⁵Wolof has no expletives. With weather verbs, a 3rd person pronominal subject occurs, and existential constructions are formed with the verb *am* ‘have’, again featuring a 3rd person clitic subject.

(17) *A'-fronting structures are not biclausal clefts*

- a. *Ngañ (woon) a (woon) taal-oon kër gi.
 Ngañ PST C_{WH} PST burn.down-PST house the.SG
 intended: 'It was Ngañ that burned down the house.'
- b. Ngañ a taal-oon kër gi.
 Ngañ C_{WH} burn.down-PST house the.SG
 'It is Ngañ that burned down the house.'

In addition to the position for A'-moved elements, Wolof also has a left-peripheral position for topicalized constituents; (18) contains a topicalized object and an A'-extracted subject. Topicalized elements are obligatorily co-indexed with clause-internal clitics (Russell 2006).

(18) *Topicalization in Wolof*

- Fas wi₁, ma-a=*(ko₁) jënd.
 horse the 1SG-C_{WH}=3SG.OBJ buy
 'The horse, it's me who bought it.' (Russell 2006: 127)

The order of the topicalized and the A'-extracted constituent is fixed: as (19a) shows, the wh-word cannot precede a topicalized constituent, it can only follow it, as in (19b). Any constituent can be topicalized in Wolof, and multiple topics can occur, as in (19c).

(19) *Topics precede 'foci'*

- a. *Kan fas wi₁ l-a=(ko₁) jënd?
 who horse the l-C_{WH}=3SG.OBJ buy
 intended: 'The horse, who bought it?'
- b. Fas wi₁ kan l-a=*(ko₁) jënd?
 horse the.PL who l-C=3SG.OBJ buy
 'The horse, who bought it?'
- c. Man₁, fas wi₂, ma₁-a=ko₂ jënd.
 1SG.STR horse the.SG 1SG-C_{WH}=3SG.OBJ buy
 'Me, the horse, it's me who bought it.'

A common property of topics is that they cannot be bare quantifiers (Rizzi 1986, 1997); this is also the case in Wolof, and will be important in diagnosing the structure of copular clauses. Example (20) shows that a bare quantifier cannot precede an A'-extracted element.

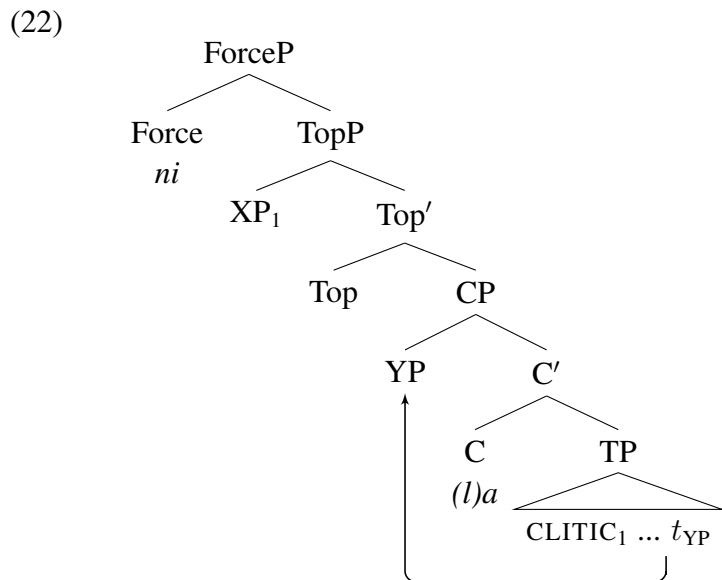
(20) *Bare quantifiers cannot be topics*

- *Kenn₁ lan l-a=∅₁ jënd?
 someone what l-C_{WH}=3SG buy
 intended: 'Someone, what did s/he buy?'

To complete the picture of the structure of the left periphery, the example in (21) shows the embedding complementizer *ni* (or *ne* for some speakers), which is distinct from the other sentence particles and can embed any finite clause with any sentence particle; in fact, the whole left periphery can be embedded in Wolof. I follow Rizzi (1997) and place the embedding complementizer in Force.

- (21) Fanta xam na= $\emptyset$ ni kër gi₁, Ngañ a=ko₁ taal.
 Fanta know C=3SG that house the, Ngañ C_{WH}=3SG.OBJ burn.down
 ‘Fanta knows that, as for the house, it’s Ngañ who burned it down.’

Based on the data presented above, I assume that the left periphery in Wolof has the structure in (22). I do not employ a Focus projection here, but label the projection to which elements A'-move as CP.



Before moving on to the structure of copular clause, I wish to briefly address the information-structural properties of A'-extraction constructions discussed above. As mentioned in section 2, constructions such as these are commonly taken to express ‘focus’. Information-structure of these clauses is not at the center of this paper, but it is important to point out several of their properties that will eventually need to be explained in light of the data on copular clauses. First, it has been clear for some time that A'-fronting of the type we are interested in here cannot be straightforwardly tied to focus. For example, both subject fronting, in (23a), and object fronting, in (23b) are felicitous as answers to a question meant to elicit a broad sentence focus, such as *Why are you angry?*. Additionally, an answer with a structure that does not involve A'-extraction and is commonly used for predicate focus, in (23c), is also felicitous.⁶

⁶Clauses as in (23c) are often called *explicative* in the descriptive literature, as they are appropriate answers to *why*-questions. They are also felicitously used in contexts eliciting predicate focus, and are generally not felicitous in out-of-the blue situations. This, however, holds only of eventive verbs; with stative verbs, they are commonly the speakers’ first choice in an out-of-the-blue context. This speaks strongly against functionalist analyses such as the one by Robert (2000, 2010), claiming that ‘focus’ is *grammaticalized* in Wolof. Already the few examples here contradict this view. It is, of course, difficult to know what ‘focus’ means in such works, as it is not formalized; for example, Robert (2010) says: ‘in my terminology, the notion of “subject focus” applies to a morphological form which prototypically indicates that the syntactic subject is the rheme. In discourse, however, focusing forms may have a variety of uses in addition to this prototypical one indicating that the lexical content of the focused constituent is the informative part of the sentence, in particular for the verb which is a syntactic constituent having the special status of predicate.’ Rheme for Robert simply means ‘new information’; see footnote 5 which shows that ‘subject focus’ can be used even when the subject is given. Robert goes on to say that *subject focus* constructions encode: (i) Identification or qualitative designation of the subject, (ii) Definition or explanation of the current situation, (iii) Exclamation with an intensification of the verb. It is difficult to understand how subject focus is “grammaticalized” given this range of meanings, or what that would even mean.

- (23) Lu tax nga mer?
 why C_{WH}.2SG angry
 ‘Why are you angry?’
- a. Sama doom a sàcc oto.
 1SG.POSS child C_{WH} steal car
 ‘MY SON stole a car.’
- b. Oto l-a sama doom sàcc.
 car l-C_{WH} 1SG.POSS child steal
 ‘My son stole a CAR.’
- c. Sama doom daf-a=∅ sàcc oto.
 1SG.POSS child do-C=3SG steal car
 ‘My son STOLE A CAR.’

Fanselow and Lenertová (2011) (see also Branan and Erlewine 2023) call this *subpart of focus fronting*, and based on German and Czech argue that the left periphery is not confined to topics and foci, and that movement to the left periphery therefore cannot be triggered or licensed by features related to information structure, but is primarily *formal fronting*.⁷

The second important property of these constructions is something we may want our analysis of A'-extraction in cleft-like constructions and copular clauses to capture. As noted by É. Kiss (1998), and further elaborated on by Horvath (2007), fronting to the preverbal position in Hungarian results in an exhaustive reading of the fronted element. This is also the case in Wolof, illustrated in the dialogue in (25). The first answer (A1) is felicitous in a situation where the market is just one of the places where one can buy dried hibiscus flowers, maybe the most convenient one. If, however, the second answer (A2) is used, where *ja ba* ‘the market’ is A'-fronted, the meaning is exhaustive: the hearer understands that the market is the only place where one can buy dried hibiscus flowers. It is not felicitous to follow up with a sentence like *You can also buy them in the store*.

- (25) *Exhaustivity in Wolof*
- Q: Fan l-a=∅ mën=a=jënd-ee bisaab?
 where l-C_{WH}=3SG can=a=buy-ee hibiscus
 ‘Where can s/he buy hibiscus?’

⁷While the data from the literature on this phenomenon, as far as I can tell, always appear to involve the movement of a subpart of a focused constituent, this is not always the case in Wolof. For example, an answer in which the subject is fronted is felicitous in Wolof even in cases when the subject is given, as in the following example (answers with object fronting and no fronting are also felicitous here).

- (24) *Context:* You meet your friends Fama and her husband Demba, who was just diagnosed with diabetes and is supposed to follow a strict diet, especially refraining from eating a lot of rice. Fama is obviously upset at Demba, so you ask her: *What did Demba do?* She answers:
- A: Demba a lekk ceeb.
 Demba C_{WH} eat rice
 ‘Demba ate rice.’

While understanding and analyzing the information-structural properties of A'-movement constructions is not one of the goals of this paper, a future analysis should be able to account for this variety in meanings, and relate them to the interpretations available in copular sentences, discussed in the following subsection.

- A1: Mën na=∅=ko jënd ja ba.
 can C=3SG=3SG.OBJ buy market the.SG.DIST.
“S/he can buy it at the market.”
- A2: Ja ba la=∅=ko mën=a=jënd-ee. #Mën na=∅=ko
 market the.SG.DIST C_{WH}=3SG=3SG.OBJ can-a-buy-ee can C=3SG=3SG.OBJ
 jënd ci *supermarché* bi itam.
 buy at store the.SG also
“It’s at the market that s/he can buy it. #She can also buy it at the store.”

Additionally, elements in Spec,CP are incompatible with focus sensitive operators ‘also’ or ‘even’; they are compatible just with ‘only’, as shown in the contrast between (26b) and (27).⁸

- (26) *tamit ‘also’ incompatible with A'-movement to Spec,CP*
- a. Fama dem na=∅. Mariam tamit dem na=∅.
 Fama leave C=3SG Mariame also leave C=3SG
“Fama left. Mariame also left.”
- b. Fama dem na=∅. *Mariam tamit mu-a (>moo) dem.
 Fama leave C=3SG Mariame also 3SG.STR-C_{WH} leave
 intended: *“Fama left. #It’s also Mariam who left.”*
- (27) *rekk ‘only’ compatible with A'-movement to Spec,CP*
- Fama dem-ul=∅. Mariam rekk mu-a (>moo) dem.
 Fama go-C.NEG=3SG Mariame only 3SG.STR-C_{WH} go
‘Fama didn’t leave. It’s only Mariame who left.’

Klecha and Martinović (2015) propose that understanding how exhaustivity arises in sentences such as these also helps us understand the fact that, as we shall see below, nominal predicates move to the same position as exhaustified constituents. I return to this in section 5.

3.2 Copular clauses

Clauses with nominal predicates involve the element (*l*)*a* with the predicate preceding it and the subject obligatorily to the left of the predicate, co-indexed with a clitic which immediately follows (*l*)*a*. Clauses as in (28a) are the most natural kind of a sentence with a nominal predicate in Wolof; it is speakers’ first choice in out-of-the-blue contexts. This sentence structure can also be used when the predicate is exhaustified, for example, as in (28b).

- (28) *Clause with a nominal predicate in Wolof*
- a. [Samba ak Idi]₁ sàcckat l-a=ñu₁.
 Samba and Idy thief l-C_{WH}=3PL
‘Samba and Idy are thieves.’
- b. CONTEXT: I am accusing Samba and Idy of all sorts of criminal activities, but the speaker corrects me that they are only thieves (and not also murderers, arsonists, etc.).

⁸A reviewer wonders if *tamit* is a verbal modifier and if (26b) is ungrammatical because of its position. *Tamit* is both a nominal and a verbal modifier, but when it is a verbal modifier, the speakers I consulted prefer to place it after the verb (*Mariam dem na tamit*). It was made clear to speakers that the interpretation in (26a) should be one in which Mariame left in addition to Fama, and not in addition to doing something else.

[Samba ak Idi]₁ sàcckat rekk l-a=ñu₁.
 Samba and Idy thief only l-C_{WH}=3PL
 ‘Samba and Idy are only thieves.’

I here reproduce evidence presented in §2 of Martinović 2023b, showing that (*l*)*a* in clauses as in (28a) is the A'-movement complementizer, that the predicate moves to Spec,CP, and that the subject is a syntactic topic, as in (29).

(29) [Top DP_{SUBJ} [CP DP_{PRED} l-a [TP SCL_{SUBJ} ... [t_{SUBJ} ... t_{PRED}]]]]

First, Wolof is an obligatory wh-movement language; wh-in situ can only be interpreted as an echo question (Dunigan 1994; Torrence 2013c). If my proposal in (29) is correct, then a predicate wh-question corresponding to the declarative clause in (28a) should have the same structure, with the wh-word occupying the position to the left of (*l*)*a*, and the subject being to its left and co-indexed with a clitic following (*l*)*a*. This is indeed what we find; left-dislocation of the subject is obligatory in this structure, as shown in (30); the subject cannot stay clause-internal, as in (30b).⁹

(30) *Copular predicate question*

- a. [Samba ak Idi]₁ lan l-a=ñu₁?
 Samba and Idy what l-C_{WH}=3PL
 ‘What are Samba and Idy?’
- b. *Lan l-a Samba ak Idi?
 what l-C_{WH} Samba and Idy

More evidence that (*l*)*a* in copular clauses is indeed the A'-movement complementizer comes from long-distance movement. Long-distance extraction in Wolof requires the complementizer (*l*)*a* in the embedded clause (Dunigan 1994; Martinović 2015, 2017), and resumptive pronouns are not grammatical in such cases, as shown in (31a). An A'-dependency into a complement clause headed by a different sentence particle, for example *na*, can be established, but it requires the presence of a resumptive pronoun, as shown in (31b) (data from Torrence 2013c); I assume that these constructions do not involve movement (Martinović 2024).

(31) *Long-distance extraction requires the complementizer la*

- a. tééré bi₁=leen=fa xale yi wax ne [CP la=a=(*ko₁) jënd]
 book C_{REL}=3PL.OBJ=LOC child the.PL say that C_{WH}=1SG=3SG.OBJ buy
 ‘the book that the children told them there that I bought’ (Torrence 2013c: 108)
- b. tééré bi₁=leen=fa xale yi wax ne [CP jënd na=ñu=*(ko₁) démb]
 book C_{REL}=3PL.OBJ=LOC child the.PL say that buy C=3PL=3SG.OBJ yesterday
 ‘the book that the children told them there that I bought it yesterday’ (Torrence 2013c: 108)

Predicates in Wolof are pronominalized with the object pronoun *ko*, as (32) shows. Wolof does

⁹As discussed at the beginning of this section, non-clitic subjects in A'-extraction constructions can be clause-internal, unlike in non-extraction clauses, which is why (30b) would not contain a clitic pronoun in addition to a non-clitic subject. In A'-constructions, they are in complementary distribution clause-internally, so a clitic occurs only in the absence of a non-clitic subject.

not have VP ellipsis, therefore the pronominalization of the predicate is obligatory.

(32) *Predicate pronominalization*

Faatu ndongo la= $\emptyset$, Musaa nekk-u(l)= $\emptyset$ =ko.
 Fatou student C=3SG.SBJ, Moussa be-NEG.C=3SG.SBJ=3SG.OBJ
 ‘Fatou is a student, Musaa isn’t (it).’

In this example a different clause structure, one involving the overt copular verb *nekk* ‘be’, is used when the pronominal predicate occurs, as negation only occurs as a verbal suffix. In this clause, there is no A'-movement; its syntax is parallel to the sentence in (10) (Martinović 2015, 2023a). As seen in (33), establishing a long-distance dependency into (32) is possible, but, just as in (31b), given that there is no A'-movement complementizer, it requires the predicate nominal to be resumed with the object clitic.

(33) *Long-distance dependency into a copular clause with resumption*

CONTEXT: Samba and Idy are involved in all kinds of criminal activities: they are murderers, they are arsonists, they are kidnappers, but my friend Moussa believes that there is one thing that they are not. I ask our mutual friend Demba:
 Lan₁ l-a Musaa xalaat na= $\emptyset$ ni Samba ak Idi nekk-ul=ñu=*(ko₁)?
 what l-C_{WH} Moussa believe C_{WH}=3SG that Samba and Idy be-NEG.C=3PL=3SG.OBJ
 ‘What does Moussa believe that Samba and Idy are not (it)?’

Long-distance extraction from the sentence in (30), on the other hand, involves no resumption, which is to be expected if (l)a is the A'-movement complementizer, and the predicate nominal can move long-distance because it already occupies Spec,CP of the embedded clause, demonstrated in (34).

(34) *Long-distance extraction from a copular clause*

CONTEXT: Moussa believes that Samba and Idy are some kind of criminals, so I ask our mutual friend Demba:
 Lan₁ l-a Musaa xalaat na= $\emptyset$ ni Samba ak Idi t₁ l-a=ñu?
 what l-C_{WH} Moussa believe C_{WH}=3SG that Samba and Idy t l-C_{WH}=3PL
 ‘What does Moussa believe that Samba and Idy are?’

Turning to the subject of copular sentences, it is syntactically a topic. First, it is obligatorily co-indexed with a clause-internal clitic, as are all topicalized DPs in the language (Russell 2006). If it is a pronoun, it can only be one from the strong series, as is expected for pronouns in the left periphery (Cardinaletti and Starke 1999; Martinović 2015). Clause-internal pronouns can only be strong if they are complements of prepositions, or if they are focused in-situ. There is no focusing involved in (35).

(35) *Copular subjects are strong pronouns*

{Ñoom₁/*ñu} sàcckat l-a=ñu₁.
 3PL.STR/3PL thief l-C_{WH}=3PL
 ‘They are thieves.’

Second, subjects in copular sentences cannot be bare quantifiers (Rizzi 1986, 1997), which we saw is true of topics in Wolof in general. In order to express the meaning in (36a), the

quantifier must be embedded inside a DP (e.g., *someone in this group*; *someone amongst us*). Clause-internal subjects can be quantifiers, as in (36b).

(36) *Copular subjects cannot be bare quantifiers*

- a. *Kenn sàcckat l-a=∅.
 someone thief l-C=3SG
 intended: ‘*Someone is a thief.*’
- b. Kër gi l-a kenn taal.
 house the.SG l-C_{WH} someone burn.down
 ‘*It’s a house that someone burned down.*’

It is also impossible to question the subject in the copular construction we are investigating here, shown in (37a), which is expected if the subject is a topic. If any element other than the predicate is extracted, a different construction is used—one with an overt copula, with the predicate nominal following it, as in (37b).

(37) *Copular subjects cannot be extracted*

- a. *Kan sàcckat l-a=∅?
 who thief l-C_{WH}=3SG
 intended: ‘*Who is the thief?*’
- b. Kan a di (> ay) sàcckat?
 who C_{WH} COP thief
 ‘*Who is a thief?*’

The contrast between copular clauses in the left periphery and those with the copula *di* drives home the point that the constituents of the former kind occupy a different layer of the clause and that such structures are verbless. (37b) is a regular clause with wh-movement of the subject and a verbal copula, *di*, which takes verbal morphology such as negation, in (38a). Negation is obligatorily a suffix on a verbal head, and cannot occur in copular clauses we have been investigating, as in (38b).¹⁰

- (38) a. Lamine a d(i)-ul jàngalekat.
 Lamine C_{WH} COP-NEG teacher
 ‘*It’s Lamine who is not a teacher.*’
- b. *Lamine jàngalekat l-a-wul=∅.
 Lamine teacher l-C_{WH}-NEG=3SG
 intended: ‘*Lamine is not a teacher*’

The position of the optional past tense morpheme (*w*)oon (Bochnak and Martinović 2019) also shows that copular constructions without a verb are built higher in the clause. The past tense morpheme is usually a suffix on the verb, but not obligatorily – it can occur in verbless clauses as well. In the copular clauses in the left periphery, it follows (*l*)a and the subject clitic, as in (39).

¹⁰I have placed the negation as a suffix on (*l*)a, which is where we would expect it to occur were (*l*)a a verbal element, but note that negation cannot occur anywhere in (38b).

- (39) *Tense morpheme in copular clauses*
 Samba ak Idi sàcc l-a=ñu woon.
 Samba and Idy thief l-C_{WH}=3PL PST
 ‘Samba and Idy were thieves.’

There is one other structural configuration in Wolof where the past tense morpheme is not affixed onto the verb. It happens in non-extraction configurations, when the verb is suffixed with negation and moves to C. The past tense morpheme remains stranded below the sentence particle and subject clitic, by hypothesis inside the TP (Torrence 2003; Martinović 2019). (For evidence that the verb is in C in (40a), see Martinović 2015, 2019, 2023a.) When there is no negation, the past tense is affixed onto the verb, as in (40b).

- (40) a. Samba ak Idi sàcc-u(l)=ñu woon oto.
 Samba and Idy steal-C.NEG=3PL PST car
 ‘Samba and Idy did not steal a car.’
 b. Samba ak Idi sàcc-oon-na=ñu oto.
 Samba and Idy steal-PST-C=3PL car
 ‘Samba and Idy stole a car.’

The reason for the different position of *oon* is not relevant here (see Torrence (2003); Martinović (2019) for different analyses), what matters is that the past tense morpheme is not necessarily affixed onto the verb, and when it is not, it follows the C-subject clitic complex. The position of the past tense in copular clauses as in (39) therefore confirms that the two nominals are above the TP domain.

In clauses with the copula *di*, (*w*)*oon* is suffixed onto the copula, as in (41a). Additionally, the position of VP-adverbials also illustrates the structural differences between a copular clause with a predicate in Spec,CP, and the one with a subject in Spec,CP: in the former, demonstrated in (41b), the adverbial must be clause-final and cannot intervene between any of the constituents, which is expected if they are all higher than the VP. In a subject extraction sentence, on the other hand, the adverbial can occur between the copula with the past tense morpheme and the nominal predicate, showing that the predicate DP in the presence of a copula is in some lower, VP-internal position.

- (41) *The position of VP adverbials in copular clauses*
 a. Lamine a d(i)-oon (**ci lu yàgg-ul**) jàngalekat (**ci lu yàgg-ul**).
 Lamine C_{WH} COP-PST in C be.long-NEG teacher in C be.long-NEG
 ‘It’s Lamine who was briefly a teacher.’
 b. Lamine (***ci lu yàgg-ul**) jàngalekat l-a=∅ woon **ci lu yàgg-ul**.
 Lamine in C be.long-NEG teacher l-C_{WH}=3SG PST in C be.long-NEG
 ‘Lamine was briefly a teacher.’

And finally, these examples also show that copular clauses in which both major constituents are in the left periphery is not in any way reduced, but is a finite clause that can contain tense morphology. This is an important point, in light of Gatchalian’s claim that in Kirundi, the copular clause that involves elements also found in A’-movement constructions, is not a full finite clause.

To summarize: clauses with nominal predicates in Wolof are finite clauses, that involve

A'-movement of the nominal predicate to Spec,CP, and topicalization of the subject.

4 The left periphery, clefting, and copular clauses in Igala

This section explores the same two constructions as section 3, focusing on Igala, a minority Yoruboid language spoken in Nigeria by about 1.6 million speakers.¹¹ Igala is a heavily understudied language, so most information presented here is novel. We first establish the structure of focus/exhaustivity constructions in Igala. We shall see that they have a more bi-clausal, cleft-like appearance, however, the ‘copular’ portion of the cleft still has some left-peripheral properties. Copular clauses themselves are very similar to those in Wolof and can be formed in the left periphery. The data presented in this section have been collected from three speakers of the Dekina variety of Igala, so the discussion and conclusions reached here are preliminary. My objective in discussing Igala is to make clearer the connection between predication and the left periphery, in using a language in which that part of the structure has mixed mono-clausal and bi-clausal characteristics. Despite the fact that the left periphery has some attributes of clefts, copular clauses formed there do not have the information-structural traits of clefts/the left periphery, the same as in Wolof. This strengthens the argument that the ability of the left periphery to host nominal predication should be at the center of the exploration of its properties.

4.1 The structure of the left periphery in Igala

Igala is an isolating, tenseless, tonal language, with a variety of aspectual morphemes. Eventive verbs are interpreted as past tense and statives as present tense. Its surface word order is SVO, though it is possibly an SOV language underlyingly, as Koopman (1984) argues is the case for some Kru languages. Example (42) illustrates a declarative, transitive Igala sentence, with progressive aspect, and (43) a sentence with a perfective auxiliary, with SOV word order. Igala has no agreement, and an overt subject—either a clitic or a non-clitic DP—is obligatory.

- | | |
|--|--|
| <p>(42) <i>SVO order in Igala</i>
 {Chìdè/î} nâ f(à) ìtébùlù.
 Chide/3SG PROG pull table
 ‘Chide/she was pulling the table.’</p> | <p>(43) <i>SOV order in Igala</i>
 {Chìdè/î} f(ū) ìtébùlù fà.
 Chide/3SG PFV table pull
 ‘Chide/she has pulled a table.’</p> |
|--|--|

Igala has several positions to the left of the subject to which it can dislocate various elements. First, it is an obligatory wh-movement language (Martinović to appear); an object question is shown in (44a). In addition to question words, Igala, as Wolof, can place other elements in the pre-subject position; (44b), with a fronted object, is a natural answer to the question *What did Ame eat?*. The sentence-final particle *yi* optionally occurs in A'-fronting constructions, regardless of which element is extracted.

- (44) *Igala object extraction*
- a. ɛ́nwú₁ Ǻmê jē t₁?
 what Ame eat
 ‘What did Ame eat?’

¹¹Eberhard, David M., Gary F. Simons, and Charles D. Fennig (eds.). 2024. *Ethnologue: Languages of the World*. Twenty-seventh edition. Dallas, Texas: SIL International. Online version: (www.ethnologue.com).

- b. òchìkāpā₁ ì jē t₁ (yí).
 rice 3SG eat SFP.FOC
'It's rice that he ate.'

In Martinović to appear, I show that this kind of fronting has the properties of A'-movement, as such constructions pass the standard movement tests, exhibiting strong and weak crossover, reconstruction for Principle A, idiom reconstruction, and island sensitivity; I do not repeat this evidence here.

Igala also has a position preceding the A'-moved element for topicalized phrases. The example in (45a) contains a topicalized subject, which is obligatorily co-indexed with a clause-internal clitic, and an A'-moved object. The topic is marked with a mid tone and the lengthening of the final vowel. Alternatively, the element *chē* must occur between the topicalized and the A'-moved element, as in (45b). *Chē* is the verb 'do', it occurs as a light verb in complex predicates, in copular clauses (as we shall see below), and between the topic and 'focus', as in (45b). I gloss it neutrally as *che*.

(45) *Subject topics in Igala*

- a. Chìdè₂-ē, ítébulù₁ ì₂ á fà t₁ yí.
 Chide table 3SG IPFV pull SFP.FOC
'As for Chide, it's a table s/he is pulling.'
- b. Chìdè₂ ch(ē) ítébulù₁ ì₂ á fà t₁ yí.
 Chide *che* table 3SG IPFV pull SFP.FOC
'As for Chide, it's a table s/he is pulling.'

Other elements can be topicalized as well; (46) shows the two options for a topicalized object.

(46) *Object topics in Igala*

- a. ítébulù₁-ū, Chìdè á f(à) ọ₁ yì.
 table Chide IPFV pull 3SG.OBJ SFP.FOC
'As for a table, it's Chide who pulled it.'
- b. ítébulù₁ ch(ē) Chìdè á fà ọ₁ yì.
 table *che* Chide IPFV pull 3SG.OBJ SFP.FOC
'As for a table, it's Chide who pulled it.'

As is the case in Wolof, topics cannot be bare quantifiers. Examples in (47) contain the element *ónè* which can have the quantifier meaning in a non-topic subject position in (47a), but not in the topic position in (47b), where it means 'the human'.¹²

(47) *Quantifiers cannot be topics*

- a. ónè á f(à) ítébulù.
 someone IPFV pull table
'Someone is pulling a table.'
- b. ónè ch(ē) ítébulù ì á fà yí.
 someone *che* table 3SG IPFV pull SFP.FOC
 **'As for someone, it's a table that s/he is pulling.'*

¹²The context given for (47b) is the following: There are various living beings in a room – a dog, a parrot, a cat, a human. The speaker is describing what each of the beings is doing.

‘As for the human, it’s a table that s/he is pulling.’

Unlike these kinds of examples in Wolof, structures as in (45b) have bi-clausal properties. First, surprisingly given the data that will follow, a cleft in which no inflectional material occurs in the ‘copular’ portion of the clause is not grammatical, as (48a) shows, with or without the relative clause complementizer *kì*, which is obligatory in relative clauses, in (49). The only way to express the intended meaning is through the structure in (48b), where only the object is A’-fronted.

(48) *Cleft with no inflectional morphology ungrammatical*

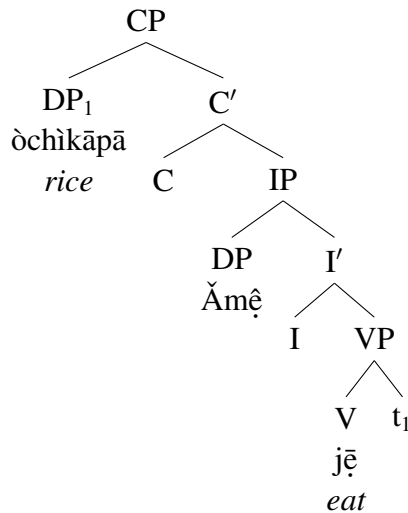
- a. *ì ch(ē) òchìkāpā (kì) Ǻmē jē yí.
 3SG *che* rice C_{REL} Ame eat SFP.FOC
 intended: ‘It is rice that Ame ate.’
- b. òchìkāpā Ǻmē jē yí.
 rice Ame eat SFP.FOC
 ‘It’s rice that Ame ate.’

(49) *Relative clause in Igala*

- [òchìkāpā *(kì) Ǻmē jē] á rínyò.
 rice C_{REL} Ame ate IPFV sweet
 ‘The rice that Ame ate is tasty.’

Note again that Igala is not a *pro*-drop language; additionally, it has no verbless clauses either (which will become clearer when we look at copular clauses), therefore I take the absence of the expletive and the copula to indicate the absence of the structure that would host them. I propose that the sentence in (48b) has the structure in (50).¹³

(50)



However, the occurrence of any inflectional material in the topic-focus part of the clause makes the cleft structure grammatical. I illustrate with aspectual morphology—the imperfective aspect *á*, and the future marker *nyá*. Both of them can occur in the ‘copular’ portion of the

¹³I omit the sentence final particle in (50). Igala has several sentence-final particles, and my preliminary observations indicate that they are situated in the C-domain, similar to such elements in Southeast Asian language. However, no work exists on sentence-final particles in Igala, and since the focus particle can also be omitted, I believe it is not crucial for the empirical picture I am painting here.

clause, and in the putative subordinate clause. Note that the only kind of a finite non-matrix clause that does not have an overt complementizer is this one: relative clauses and subjunctive-like non-finite clauses obligatorily occur with the complementizer *kí* and *kí̄*, respectively, and finite clauses embedded under bridge verbs are headed by the complementizer *kàkíní*. None of these elements can occur in (51).

- (51) ì á nyá ch(ē) Àládi Āmē á nyá lí yì.
 3SG IPFV FUT *che* Aladi Ame IPFV FUT see SFP.FOC
 ‘It will be Aladi that Ame will see.’

These constructions look like clefts of the English type, but the ‘copular’ portion of the cleft retains some properties of the left periphery. For example, a topicalized constituent can occur instead of the expletive, as in (45b); note that subjects are otherwise never freely omitted. (52) shows the occurrence of aspectual morphology between the topic, in this case either the subject or the object, and *chē*. Note that the meaning is still that of a cleft; (52a) does not have the nonsensical meaning *Chide will be the table that she will pull*, nor does (52b) mean *The table will be Chide who will pull it*.

- (52) *Aspectual morphology in the left periphery in Igala*
 a. Chìdè₁ á nyá ch(ē) ìtébùlù ì₁ á nyá fà yì.
 Chide IPFV FUT *che* table 3SG IPFV FUT pull SFP.FOC
 ‘As for Chide, it will be a table that she will pull.’
 b. ìtébùlù₁ á nyá ch(ē) ìChìdè á nyá fà ó₁ yì.
 table IPFV FUT *che* Chide IPFV FUT pull 3SG.OBJ SFP.FOC
 ‘As for the table, it will be Chide who will pull it.’

Even more interesting is that the topic can carry inflectional morphology, specifically negation. First, negation in Igala is bipartite, consisting of a high tone which docks onto the last vowel of the subject, and a sentence-final particle, as shown in (53) (Chaperon to appear).¹⁴

- (53) *Negation in Igala*
 Chìdě f(à) ìtébùlù ní.
 Chide.NEG pull table SFP.NEG
 ‘Chide didn’t pull a table.’

High tone negation can occur on the expletive element of the cleft, as in (54a). Negation in the lower clause is also possible, shown in (54b). The form of negation, *má*, which occurs in A’-constructions, confirms that structures we have been investigating involve A’-extraction, and are not some kind of a paratactic constructions.

- (54) a. í ch(ē) òchìkāpā Chìdè má jē yí ní
 3SG.NEG *che* rice Chide NEG eat SFP.FOC SFP.NEG
 ‘It isn’t rice that Chide didn’t eat.’
 b. Chìdě ch(ē) òchìkāpā ì má jē yí ní
 Chide.NEG *che* rice 3SG NEG eat SFP.FOC NEG

¹⁴As many Niger-Congo languages, negation has two forms in Igala: the high tone occurs in declarative clauses, and in A’-movement constructions and nominalizations, the morpheme *má* occurs. The sentence-final particle is always obligatory.

‘As for Chide, it isn’t rice that she didn’t eat.’

The data presented in this section are preliminary, and there is still much work to be done to understand the syntactic properties of clefts and the left periphery in Igala. For the purposes of this paper, I will assume that Igala has both clefts and structures with a Wolof-style left periphery. Clearly the divide between the two is not clean, and it is likely that constructions with topics and foci are being grammaticalized into cleft constructions (or the other way around), but this process is not completed. This results in a few peculiar properties of Igala clefts: that they are not grammatical if no inflectional morphology occurs, but are if such morphology is present, that the topic can occur in place of the expletive and carry inflectional morphology, and that the subordinate clause is the only such clause not containing an overt complementizer. This suggests that the position of the expletive is not, in fact, Spec,IP of a regular clause, but that it still has the syntactic properties of a topic. The data on copular sentences will bring even more variation into play, as we shall see that there the part of the clause that I will argue is the left periphery looks like what we saw in Wolof – the subject and the predicate exhibit the syntactic properties of topics and ‘foci’, but do not obligatorily exhibit their information-structural properties.

For the sake of completeness, I briefly show that A'-extraction in Igala has the same basic properties as in Wolof. First, it is also not directly tied to focusing, shown by examples in (55), which illustrate subpart of focus fronting and parallel those from Wolof given in (23): (55a) involves no extraction, in (55b) the subject is extracted, and in (55c) the object.

- (55) *Subpart of focus fronting*
- énwú chí k(ì) ẹ́ w(ó) ẹ́dò?
 what cause C 2SG.IPFV foam heart
 ‘Why are you angry?’
- a. ọmā mī j(í) ímótò.
 child 1SG.POSS steal car
 ‘My son stole a car.’
- b. ọmā mī j(í) ímótò yí.
 child my steal car SFP.FOC
 ‘MY SON stole a car.’
- c. ímótò ọmā mī jí yì.
 car child my steal SFP.FOC
 ‘My son stole a CAR.’

Additionally, A'-fronting in Igala also yields an exhaustive reading of the fronted element; for some speakers the exhaustivity is more prominent in the presence of the sentence-final particle *yì*. The answer in (56b), in response to (56a), indicates that the market is the only place where one can buy locust beans; it cannot be followed up with a sentence suggesting another place where one can buy them. If the market is just one of the places where you can buy locust beans, then a regular sentence without A'-fronting is used, as in (56c).

- (56) *A'-extraction involves exhaustivity*
- a. ūgbò ù nēkē l(à) ūgbá?
 where 1SG can buy locust.bean

‘Where can I buy locust beans?’

- b. éf(û) ājá è nēkē là yí. #è nwọ nēkē là éf(û) īstō.
 at market 2SG can buy SFP.FOC 2SG also can buy at store
 ‘It’s at the market that you can buy them. #You can also buy them at the store.’
- c. è nēkē là éf(û) ājá. è nwọ nēkē là éf(û) īstō.
 2SG can buy at market 2SG also can buy at store
 You can buy them at the market. You can also buy them at the store.

We now move on to the copular clauses in Igala, which, I shall argue, are built in the left periphery, just like Wolof copular clauses.

4.2 Copular clauses

Copular clauses in Igala are strikingly similar to those in Wolof. A predication sentence with a nominal predicate is given in (57). I also provide a context in which this sentence occurs naturally.

- (57) CONTEXT: At a party, I am introducing my friend Aladi to you. *This is my friend Aladi, we grew up together.*
 Ālādì₁ ch(ē) ītíchà ì₁ chē.
 Aladi *che* teacher 3SG *che*
 ‘Aladi is a teacher.’

This construction can also be used to answer a predicate question (*What is Aladi?*, as in, what is her profession), and in contexts in which the predicate is contrastively focused.

In the same context as in (57), speakers also accept the sentence in (58), though a common comment is that it is a bit curt, “not rounded up”.

- (58) Ālādì ch(ē) ītíchà.
 Aladi *che* teacher
 ‘Aladi is a teacher.’

In (58), the subject and the predicate appear to occupy the standard positions inside the IP, as I will show below. My focus here are sentences as in (57), which I argue have the syntax in (59). I will show that the subject nominal has the properties of a syntactic topic, and the predicate nominal A'-moves to Spec,CP. And while, as noted above, this kind of a copular sentence can be used when the predicate nominal is focused, the key observation is that it occurs in contexts in which that is not the case.

- (59) [_{TopP} DP_{SUBJ} *che* [_{CP} DP_{PRED} [_{IP} SCL_{SUBJ} *che* [_{t_{SUBJ}} ... _{t_{PRED}}]]]]

First, I present the same kinds of data we saw for Wolof to confirm that the subject and the predicate are in left-peripheral positions. Given that Igala is an obligatory wh-movement language, we should be able to felicitously substitute the predicate with a wh-word without changing the remaining structure, as we did in Wolof. This yields a grammatical predicate question, as shown in (60).

- (60) *Predicate question*
 Āládì ch(ē) ēnwū ì chē?
 Aladi *che* what 3SG *che*
What is Aladi? (what is her profession)

In the same vein, if the subject is a topic in this structure, it should not be able to be a question word; this is also born out, as shown in (61a). The question is felicitous if ì *chē* is left out, as expected if, in that case, the subject and predicate are in their canonical clause-internal positions, and the subject can move to Spec,CP (which is here string-vacuous, since Igala does not have an overt A'-movement complementizer), as in (61b).

- (61) *Subject question*
 a. *ēñē₁ ch(ē) ītíchà ì₁ chē?
 who *che* teacher 3SG *che*
 intended: ‘Who is a teacher?’
 b. ēñē ch(ē) ītíchà?
 who *che* teacher
 ‘Who is a teacher?’

Just as in Wolof and many other languages that have strong and weak/clitic pronouns, left-peripheral positions can only be occupied by strong pronouns. This is the case in copular clauses that we are exploring here, as shown in (62): the clitic ì is impossible, only the strong pronoun ònwù is allowed. As in Wolof, (62b) does not require the subject to be focused, or for it to be a topic, which is otherwise a requirement for the usage of a strong pronoun for the subject.

- (62) *Subjects are strong pronouns*
 a. *ì₁ ch(ē) ītíchà ì₁ chē.
 3SG *che* teacher 3SG *che*
 intended: *She is a teacher.*
 b. ònwù₁ ch(e) ītíchà ì₁ chē.
 3SG.STR *che* teacher 3SG *che*
 ‘*She is a teacher.*’

Additionally, just like we saw is the case for topicalized constituents in (47), the element ónè cannot be interpreted as a quantifier in the sentence in (63); it can only mean ‘the human’ (in a similar context as (47b)).

- (63) ónè₁ ch(ē) ītíchà ì₁ chē.
 human *che* teacher 3SG *che*
 *‘Someone is a teacher,’
 ‘The human is a teacher.’

Aspectual marking in this kind of a clause is located inside the constituent marked here as IP, illustrated with the imperfective aspect and the future marker *nyá*, in (64), not in the part of the clause that I argue constitutes the left periphery.

- (64) *Aspectual morphology is found inside the IP*
- a. [Ālādī₁ ch(e) ītíchà [ī₁ á nyá chē]]
 Aladi che teacher 3SG IPFV FUT che
 ‘Aladi will be a teacher.’
- b. *Ālādī₁ á nyá ch(ē) ītíchà ì₁ chē
 Aladi IPFV FUT che teacher 3SG che
 intended: ‘Aladi will be a teacher.’

Given the data we have seen in clefts, we may expect that these structures should show the same properties; that is, that they should be able to be interpreted as clefts. (64b) might then be infelicitous because of the mismatch in the aspectual marking in the matrix and embedded clause (i.e., it might mean something like *It will be a teacher that Aladi is*, which may be the cause of the ungrammaticality). Indeed, when the aspectual marking is identical in both parts of the clause, such examples improve, but they do not have the meaning of a cleft, as (65) shows.

- (65) Ālādī₁ á nyá ch(ē) ītíchà ì á nyá chē.
 Aladi IPFV FUT che teacher 3SG IPFV FUT che
 ‘It will eventually be the case that Aladi will be a teacher.’
 ‘It might be that Aladi will be a teacher.’¹⁵

The speaker who offered the first interpretation also accepted (64b), under the interpretation *It will eventually be discovered that Aladi is a teacher after all*. What is interesting is that this is not a case of regular embedding in Igala. For example, the embedding complementizer *kàkíní* cannot be added anywhere in (65). When asked to give a sentence that means *It will be the case that Aladi will be a teacher*, speakers naturally offer the sentence in (66a). It is possible to also A'-move the predicate in the embedded clause in a focusing context, as in (66b), but crucially, the subject *Aladi* cannot replace the expletive in the matrix clause. These are true bi-clausal structures.

- (66) a. ì á nyá chē *(kàkíní) Ālādī á nyá ch(ē) ītíchà.
 3SG IPFV FUT che that Aladi IPFV FUT che teacher
 ‘It will be the case that Aladi will be a teacher.’
- b. ì á nyá chē *(kàkíní) ītíchà₁ Ālādī á nyá ch(ē) t₁.
 3SG IPFV FUT che that teacher Aladi IPFV FUT che
 ‘It will be the case that Aladi will be a TEACHER.’

The data on the left periphery in Igala are more complex than what we saw in Wolof, and more research is required for them to be well understood. ‘Focus’ constructions have some bi-clausal properties, but do not appear to be fully grammaticalized into clefts. Copular clauses, however, are quite similar to the ones in Wolof, with the subject and the predicate begin able to occupy positions in the left periphery. Crucially, in such constructions, the information-structural properties usually associated with the topic and ‘focus’ positions are not obligatorily present, but the constituents occupying those positions have the same morphosyntactic properties in both types of clauses: the element which is by hypothesis in the topic position exhibits

¹⁵The speaker who offered this meaning found the sentence somewhat degraded, and did not find the meaning *It will eventually be the case that Aladi will be a teacher* to be the best one. The speaker gave this meaning instead and preferred to add *àyô* ‘maybe’ to the end of the sentence.

a typical property of topics (e.g., it cannot be a quantifier), and the element which is by hypothesis in Spec,CP exhibits properties of an A'-moved element. This drives home the point that the left periphery is not the locus of information-structural features, but that those properties of utterances must be derived in another way.

In the following section, I explore what the evidence presented thus far means for the architecture of the clause. I do not offer definitive answers in this paper, but hope to identify the questions I believe arise from this, and how we may go about answering them.

5 Discussion

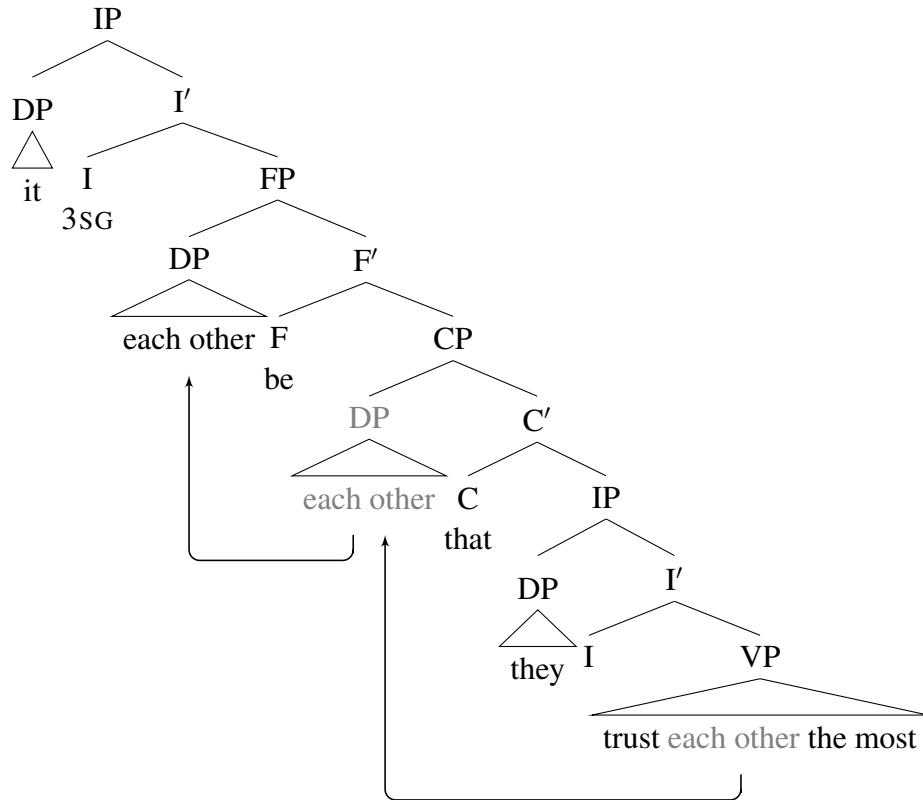
In the previous two sections we investigated the properties of the left periphery/cleft constructions in Wolof and Igala. The descriptive literature and the literature on grammaticalization have long noted that it is common for topic and focus markers to be reanalyzed into copulas (e.g., Li and Thompson 1977; Heine and Reh 1984), and this tendency is confirmed by the languages we discussed here. Both Wolof and Igala A'-extract nominals for exhaustivity, and both languages form clauses with nominal predicates in the left periphery, with the subject being a syntactic topic, and the predicate A'-moving to Spec,CP. Additionally, the Igala left periphery has clausal characteristics: the same element that functions as a copula clause-internally is also found in the left periphery, between the topic and the A'-extracted element, and the left periphery can host inflectional material. If the left periphery is being grammaticalized into an independent clause in Igala, the process does not appear to be complete.

Igala appears to be between languages like Wolof and languages like English, where we have no doubt that structures with A'-movement for focus/exhaustivity are embedded under copular verbs in bi-clausal clefts. É. Kiss (1999) argues that the cleft construction involves movement of the clefted constituent to the specifier of a Focus Phrase (which Brody 1990 proposed for the Hungarian preverbal 'focus' position), which is then selected by a copula, demonstrated in (67). Note that, in English, A'-movement of non-wh XPs for exhaustivity (**each other they trust the most*) is not licit when the clause is not embedded under a copula; this is not the case in Igala.

(67) É. Kiss's (1999) analysis of English clefts

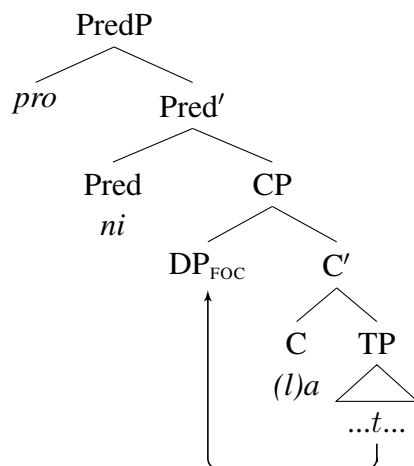
- a. It is each other that they trust the most.

b.



Another language that is somewhere in-between Wolof and English is Kirundi, whose clefts also have bi-clausal properties in that the part of the clause following the A'-moved constituent has morphosyntactic characteristics of an embedded clause. Interestingly, the copular part of the clause does not exhibit clausal properties. Gatchalian (2023, to appear) proposes that the left periphery has become an independent domain, albeit a very reduced one, without any verbal element and without an extended projection, as in (68).

(68)



If these analyses of Wolof, Igala, Kirundi, and English are on the right track, they are potentially capturing different stages in the grammaticalization of the left periphery into a copular clause (or the other way around; I have nothing to say about the direction of this change). According to this view, the left periphery would be a special structure used for what we may call ‘basic predication’, though we would, of course, need to precisely define what we mean by

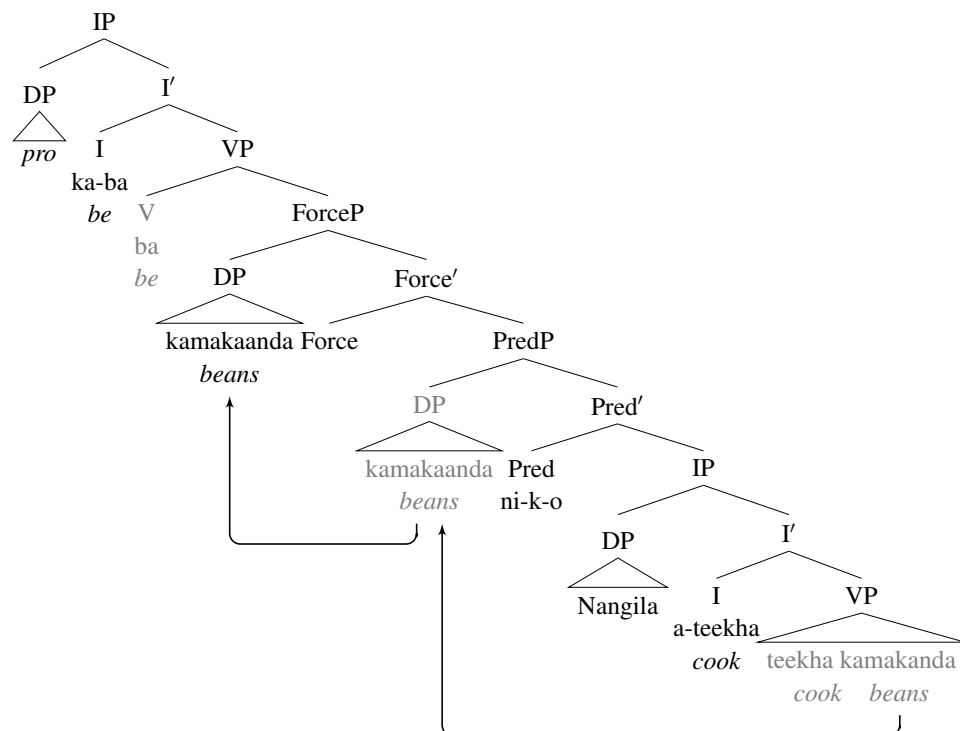
this, both syntactically and semantically. In order to shed some light on this puzzle, I focused on precisely those languages where the left periphery hosts nominal predicates. I believe the patterns discussed here suggest that understanding the relationship between movement for exhaustivity and nominal predication is crucial for a proper characterization of the left periphery and its position in the architecture of the clause.

Various works have touched on the intuition that what A'-moves in the structures at issue is some sort of a predicate. For example, Wasike (2006) proposes that the complementizer-like element *ni* (see section 2) is a head in the left periphery in Lubukusu, which he terms “Pred”, but he leaves the identity of that head vague (“Our labels, particularly PredP, are neutral and can generalize across different construction types.”; p. 142). While he argues that Lubukusu clefts are bi-clausal, this seems to be an attempt to capture the intuition that what we find A'-moving has some properties of a predicate, however, without committing to a particular analysis for Pred, our understanding of these structures is not really improved. Wasike proposes the structure for a Lubukusu cleft as in (69). The clefted constituent A'-moves to Spec,PredP (and then continues to Spec,ForceP for reasons not relevant here), and then that entire clause is the complement of the copula in the higher clause.¹⁶ The lighter coloured constituents in the tree are traces of movement.

(69) *Object cleft in Lubukusu*

- a. Ka-ba ka-ma-kaanda ni-k-o Nangila a-a-tekha.
 6-be PP-6-beans NI-6-PRON 1Nangila 1-PST-cook
 ‘It was beans that Nangila cooked.’ (Wasike 2006: 25)

b.



É. Kiss (2006) also addresses the intuition that some kinds of A'-movement to the left periphery involve predication. The Hungarian preverbal position famously hosts exhaustified con-

¹⁶The element *ni-ko-* is syntactically complex, consisting of several functional projections. This is not relevant here and is omitted from the tree for simplicity.

stituents, but also verbal particles, and nominal predicates. She argues that Hungarian does not have a FocusP projection preverbally, contra Brody (1990, 1995) and much subsequent work (including, e.g., É. Kiss 2002), but a PredP projection, which can be filled by verbal particles that function as secondary predicates, or alternatively by an A'-moved DP. The key observation that É. Kiss wants to capture is that the element in Spec,PredP is 'interpreted as a predicate'. This observation is not formalized, however; É. Kiss (2006) states it only descriptively:

“A preverbal constituent, whether simple or complex, occupies Spec,PredP, where it receives a predicate interpretation. As shown by Higgins (1973), a nominal can function as a predication, identificational, or specificational predicate. Any type of noun phrase can express specification. Predication (in the narrow sense), on the other hand, can only be expressed by a bare nominal or a nonspecific indefinite noun phrase (or by an adjective phrase).” (p.187)

As É. Kiss notes, this intuition about different kinds of predicates goes back to Higgins's (1973) influential work on specificational pseudoclefts, but she does not go much beyond that. We do not know what it means, formally, to be a predication or a specificational predicate, and consequently this does not tell us what it is about languages such as Hungarian, Hausa, Wolof, or Igala, that have A'-positions in the left periphery to which they move DPs which are then interpreted either as exhaustified, or are nominal predicates in a copular sentence.

Klecha and Martinović (2015) attempt to provide a formal, uniform account of nominal predication and exhaustive identification—Kiss's *predicational* predication and *specificational* predication—in Wolof. They analyze the complementizer (*l*)*a* as having a semantics whereby the unique individual satisfying the property denoted by its complement (the TP containing the trace of movement) has the property denoted by its specifier. The denotation of (*l*)*a* is shown in (70):

$$(70) \quad \llbracket (l)a_n \rrbracket = \lambda p \lambda Q \lambda g [Q(*\iota x [p(g^{x/n})])](g)]$$

What this means is that (*l*)*a* binds the trace of movement (as in Sternefeld 2001; Kobele 2010; Kennedy 2014) to its specifier by an iota operator. This semantics requires the element in the specifier to be a property (i.e., of type $\langle e, t \rangle$), so if the specifier is occupied by a referential DP (of type $\langle e \rangle$), it must be type-shifted into a property. The analysis translates the sentences in (71a) and (72a) into (71b) and (72b):

- (71) a. Musaa l-a=ñu gis.
Moussa l-C_{WH}=3PL see
'It's Moussa who they saw.'
b. The unique individual they saw has the property of being Moussa.
- (72) a. Jangalëkat l-a=ñu.
teacher l-C_{WH}=3PL
'They are teachers.'
b. The unique individual identical to them has the property of being a teacher.

The exhaustivity, which is imparted in (71a) is neutralized in (72a), because the property being exhaustified (the property of being a plurality identical to them) is already a singleton. For details, see the original paper. This analysis gets us a step closer to capturing the nature of

exhaustivity, but this is achieved through specific semantics given to the A'-attracting head (here, the complementizer (*la*)), without it being related to any particular property of the left periphery.

There are several additional properties of predication in the left periphery that indicate that the key relationship is the one between the topic/subject and 'focus'/predicate. First, in both Wolof and Igala the clause-internal subject (i.e., the subject below C) is a pronoun, and the non-pronominal subject is topicalized. In Wolof, this is obligatory—if the predicate is A'-extracted, the subject must be dislocated and resumed, as shown by . Having a copula does not help (recall that a copula is necessary when both constituents are not in the left periphery, which occurs when a constituent other than the predicate is A'-extracted; see (37b)).

(73) *Subject must be left-dislocated if the predicate A'-moves*

*Sàcckat l-a Musaa (di).

thief l-C_{WH} Moussa COP

intended: 'Moussa is a thief.'

In Igala, such a sentence is not ungrammatical, but it results in exhaustive reading of the predicate, as in (74).

(74) ítíchà Āládì chē.

teacher Aladi COP

'It's a teacher that Aladi is.' (as opposed to having a different profession)

The obligatory left-dislocation of the subject therefore appears to be an important property of predication in the left periphery that an analysis should capture. Klecha and Martinović (2015) can account for this, but based on a technicality. Namely, when the predicative nominal moves to Spec,CP, its trace is of type $\langle e \rangle$, which should derive a type mismatch if the other DP in a predication clause is also of type $\langle e \rangle$. They therefore stipulate that pronouns are of type $\langle e, t \rangle$, which avoids the type mismatch (see the original work for details), and forces a non-pronominal subject to be dislocated (since this is how the grammar can achieve co-indexation with the clause-internal pronominal subject). Clearly this cannot account for Igala, where the sentence with a non-pronominal, non-dislocated subject is grammatical, as in (74), it just has a different interpretation. The crucial relationship where predication is established is between the subject in Spec,TopP and the predicate in Spec,CP. Klecha and Martinović's (2015) analysis does not capture this.

Another piece of data confirms the importance of understanding the relationship between the subject and the predicate, and it is again shared between Wolof and Igala: equative sentences cannot be formed in the left periphery in either language. In Wolof, the sentence in (75a) is grammatical only if we are not equating the individual known as Clark Kent with the individual known as Superman, but if we are trying to say that the individual Clark Kent has some characteristics of the individual known as Superman. A clearer example of this is the one in (76), which can only mean that Demba sings as well or as Youssou N'Dour, or in a manner similar to him (or has some other, very salient property of Youssou N'Dour). This shows that the element in Spec,CP must be predicated of the topicalized element, and there appears to be no predication relationship in equatives. The only way to express identity between two individuals in Wolof is by A'-moving the higher of the two DPs to Spec,CP, and using a copula, as in (75b).

(75) *Equatives in Wolof*

- a. *Clark Kent Superman l-a=∅.
Clark Kent Superman l-C_{WH}=3SG
intended: ‘Clark Kent is Superman.’
- b. Clark Kent a di (>ay) Superman.
Clark Kent C_{WH} COP Superman
‘Clark Kent is Superman.’

(76) *Referential DPs in Spec,CP are interpreted as properties*

- Musaa Yusu Nduur l-a=∅.
Moussa Youssou N’Dour l-C_{WH}=3SG
meaning: *Moussa sings as well as/similarly to Youssou N’Dour.*

Similarly, equatives in Igala cannot be formed in the left periphery. As shown in (77), they are grammatical when the two DPs are clause-internal.

(77) *Equatives in Igala*

- a. *Clark Kent ch(e) īSuperman ì chē.
Clark Kent *che* Superman 3SG *che*
intended: ‘Clark Kent is Superman.’
- b. Clark Kent ch(e) īSuperman.
Clark Kent *che* Superman
‘Clark Kent is Superman.’

It is important to note that the problem is not in having a referential DP in Spec,CP in such sentences, or two definite DPs occupying the Spec,TopP and Spec,CP, but the relationship they establish with each other. For example, specificational sentences like (78)–(79) are grammatical in the left periphery in both languages.¹⁷

(78) *Specificational sentence in Wolof*

- Waaykat bi Yusu Nduur l-a=∅.
singer the.SG Youssou N’Dour l-C_{WH}=3SG
‘The singer is Youssou N’Dour.’

(79) *Specificational sentence in Igala*

- [ɛnē k(i) ì dù] ch(e) Àládi ì chē.
person C_{REL} 3SG win *che* Aladi 3SG *che*
The winner is Aladi.

An additional quirk of predication in the left periphery, shown in (80b), is that definite descriptions are banned from Spec,CP in copular clauses, in both Wolof and Igala:

¹⁷Even though the specificational subject has been argued to be a predicate, there exists solid evidence against this. Romero (2005) and Arregi et al. (2021) argue that they are individual concepts—descriptions that determine different referents at different worlds and/or times. In that case, the referential expression in Spec,CP is a property of the individual concept (i.e., it is predicated of it)—it picks out the referent in a particular world/time.

(80) *Definite descriptions banned from Spec,CP in copular clauses*

- a. *Musaa sàcc bi l-a=∅
 Moussa thief the.SG l-C_{WH}=3SG
Intended: ‘Moussa is the thief.’
- b. *Ālādì₁ ch(e) òtíchà lé ò₁ chē
 Aladi che teacher the 3SG che
Intended: ‘Aladi is the teacher.’

That this is really a restriction on what can be a predicate in these kinds of structures is confirmed by Ch’ol, which has a designated predicate position, in which it places all predicates: verbal, prepositional, and nominal. Coon and Martinović (2023) show that definite descriptions are banned from that position. A similar restriction is found in Scottish Gaelic (Adger and Ramchand 2003). Whatever strategy enables definite descriptions to function as predicates otherwise (or, to put it more neutrally, to occupy apparent predicate positions), is not available in the left periphery.

The restrictions discussed above in Wolof and in Igala are present only in copular clauses; otherwise, any kind of a DP can A’-move to Spec,CP, and be preceded by a topic. The relationship between the left-dislocated/topicalized element and the A’-extracted element in Spec,CP (often informally referred to as the ‘topic-comment structure’) therefore needs to be captured by the analysis that aims at unifying fronting for exhaustivity and nominal predication.

In this section, I have laid out previous analysis that have in some way touched on the central issue discussed in this paper: the fact that the left periphery in many languages is the locus of nominal predication. I have not proposed a new analysis here, but I hope to have laid out the issues that I believe a thorough analysis of the left periphery should capture, and proposed some directions for their further exploration. I have here not discussed languages which do appear to have a Rizgian left periphery, but which has not been reported to exhibit the properties discussed here, such as Italian. Ultimately, we need to understand whether we are dealing with syntactically universal layers of structure, with the variation being in other parts of the grammar (for example, the semantics of some of the left-peripheral heads), or if the left peripheries in languages like Italian, and the ones in languages like Wolof, are actually not akin to one another.

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